

WTW158 Calculus

8

Class Test 4 (Tu)

17 - 21 April 2023

Time: 15 min

Surname:

Memo

Initials:

Student number:

Group:

Use a pen to complete the test. Show all your steps and simplify your answers. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [1 mark]

Determine the infinite limit: $\lim_{x \rightarrow 1^+} \ln(\sqrt{x} - 1)$.

$$\lim_{x \rightarrow 1^+} \ln(\sqrt{x} - 1) = -\infty$$

QUESTION 2 [2 marks]

Find the limit, if it exists: $\lim_{x \rightarrow -2} \frac{2 - |x|}{2 + x}$.

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

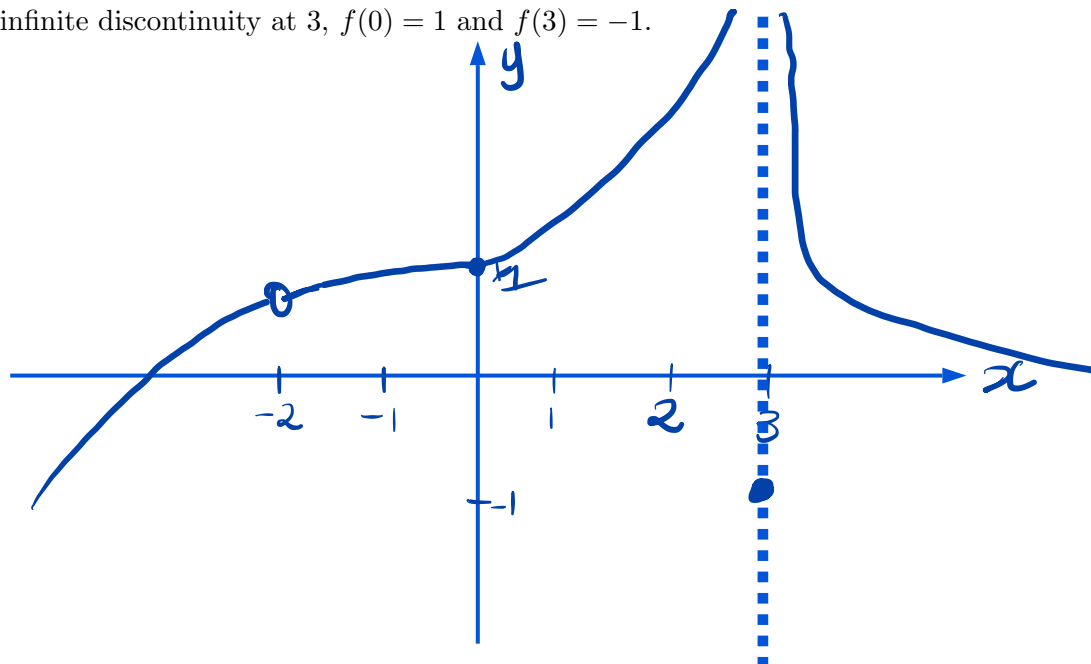
then $\lim_{x \rightarrow -2} \frac{2 - |x|}{2 + x}$

$$= \lim_{x \rightarrow -2} \frac{2 + x}{2 + x}$$

$$= \lim_{x \rightarrow -2} 1 = 1$$

QUESTION 3 [2 marks]

Sketch the graph of a function f that is defined on $\mathbb{R}$ and continuous, except for a removable discontinuity at -2 and an infinite discontinuity at 3 , $f(0) = 1$ and $f(3) = -1$.

**QUESTION 4 [3 marks]**

Sketch the graph of an example of a function f that satisfies all of the given conditions: $\lim_{x \rightarrow -\infty} f(x) = -\infty$, $\lim_{x \rightarrow \infty} f(x) = \infty$, $\lim_{x \rightarrow -2^-} f(x) = \infty$, $\lim_{x \rightarrow -2^+} f(x) = -\infty$, $\lim_{x \rightarrow 2^-} f(x) = -\infty$, $\lim_{x \rightarrow 2^+} f(x) = \infty$ and $f(0) = 1$.

